Tables Used

- dim_Products
- dim_Geography
- dim_Date
- dim_Shipping
- Fact_orders

Relationships

- Fact_orders.Product_id – dim_Product.Product_id
- Fact_orders.Geo_id – dim_Geography.Geo_id
- Fact_orders.Order_Date – dim_Date.Date
- Fact_orders.Shipment_id – dim_shipping.shipment_id

Key Calculations / Measures

- **Total Orders** – COUNTROWS(fact_orders)
- **Total Revenue** – SUM(fact_orders[Revenue_USD])
- **Total Profit** – SUM(fact_orders[Profit_USD])
- **Total Shipping Cost** – SUM(fact_orders[ShippingCost_USD])
- **Profit Margin %** – DIVIDE([Total Profit], [Total Revenue])

Notes

The date fields in the raw dataset (such as Order_Date, Expected_Delivery_Date, and Actual_Delivery_Date) were ensured to be in proper date format to support accurate time-based analysis within Power BI.

Dimension tables (dim_product, dim_geography, dim_shipping, and Dim_Date) were created by extracting distinct values from the main dataset. This approach reduces redundancy, ensures data normalization, and improves model performance.

A **star schema model** was implemented, with fact_orders as the central fact table connected to multiple dimension tables. This structure enhances query performance and enables efficient filtering and aggregation.

Relationships between fact and dimension tables were established using key columns (such as product_id, geo_id, and Shipment_ID) to maintain referential integrity and ensure accurate data analysis.

Several **calculated columns** were created, such as:

- Delivery Status (using shipment status)
- Delivery Days (difference between order and delivery date)
- Shipping Mode (from shipping dimension)

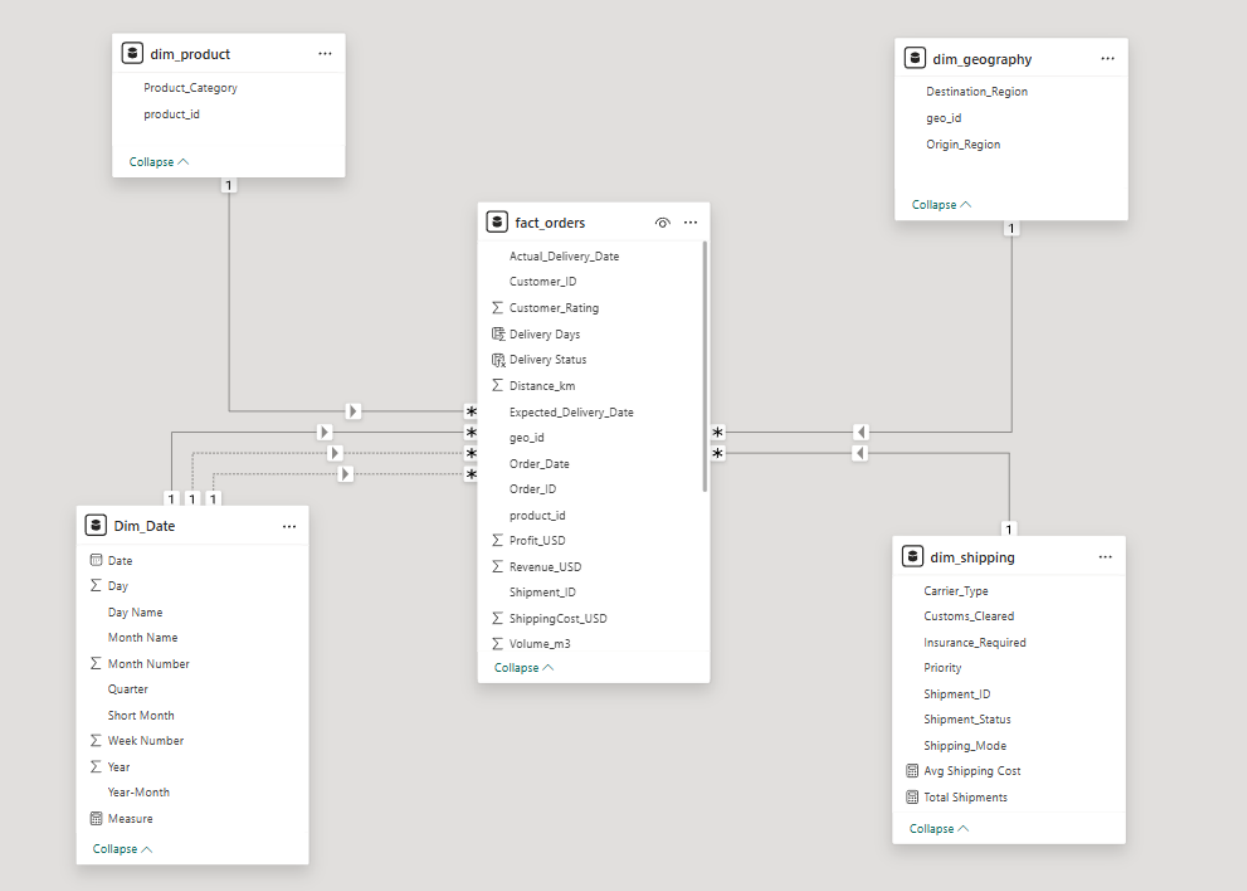
These columns support deeper operational and performance analysis.

Key performance measures (KPIs) such as **Total Revenue, Total Profit, Shipping Cost, Profit Margin, and Delay Rate** were created using DAX to provide business insights.

The dataset represents **historical logistics operations**, and the analysis focuses on identifying trends, performance patterns, and efficiency metrics rather than real-time tracking.

Customer-related fields such as Customer_Rating are used to evaluate service quality and customer satisfaction across regions and product categories.

Diagram


dim_product
Product_Category
product_id
Collapse
dim_geography
Destination_Region
geo_id
Origin_Region
Collapse
fact_orders
Actual_Delivery_Date
Customer_ID
Customer_Rating
Delivery Days
Delivery Status
Distance_km
Expected_Delivery_Date
geo_id
Order_Date
Order_ID
product_id
Profit_USD
Revenue_USD
Shipment_ID
ShippingCost_USD
Volume_m3
Collapse
Dim_Date
Date
Day
Day Name
Month Name
Month Number
Quarter
Short Month
Week Number
Year
Year-Month
Measure
Collapse
dim_shipping
Carrier_Type
Customs_Cleared
Insurance_Required
Priority
Shipment_ID
Shipment_Status
Shipping_Mode
Avg Shipping Cost
Total Shipments
Collapse